

Working Life Balance Dashboard

1. Project Overview

The **Working Life Balance Dashboard** is an interactive data analytics project designed to study the relationship between **daily lifestyle habits** and **lifespan (age at death)**. The project focuses on how individuals allocate their **24-hour day** across key activities such as **work, sleep, exercise, and rest**, and how these choices influence longevity.

This analysis is based on a **synthetic dataset of 10,000 individuals**, created to reflect realistic lifestyle patterns along with some extreme cases. The dataset enables meaningful analysis of **work-life balance, health behavior, and long-term outcomes**.

The dashboard allows users to explore:

- The impact of **working hours** on lifespan
- The role of **sleep and exercise** in healthy living
- **Occupation-wise** and **gender-wise** longevity patterns
- Differences in **daily time allocation** across professions

The project is developed using:

- **Python** for data cleaning and preprocessing
- **SQL** for structured data analysis
- **Power BI** for interactive visualization and insights

The final output is a **dynamic and user-friendly dashboard** that helps stakeholders, clients, or decision-makers easily understand how lifestyle balance contributes to long-term health and longevity.

2. Dataset Summary

The dataset used in this project is a **synthetic lifestyle and longevity dataset** consisting of **10,000 individual records**. It is designed to simulate realistic human behavior related to **daily time allocation, occupation, and lifespan**, making it suitable for analytical and visualization-based studies.

Each record in the dataset represents one individual and includes demographic, professional, and lifestyle-related attributes. The data ensures that daily activities such as **work, sleep, rest, and exercise collectively sum to approximately 24 hours**, reflecting real-world daily routines.

Key Features of the Dataset:

- **Age at Death:** Represents the lifespan of individuals
- **Occupation Type:** Categorizes individuals by profession
- **Daily Work Hours:** Average time spent working per day
- **Daily Sleep Hours:** Average sleep duration per day
- **Daily Exercise Hours:** Time spent on physical activity
- **Daily Rest Hours:** Remaining non-working and non-sleep time
- **Gender:** Demographic classification

The dataset also includes **outliers**, such as individuals with extremely long working hours or minimal rest, to support deeper analysis of **non-linear relationships** and **diminishing health returns**.

Purpose of the Dataset:

- To analyze the **impact of lifestyle choices on longevity**
- To study **work-life balance patterns across occupations**
- To support **regression-style and comparative analysis**
- To enable **interactive dashboard exploration**

Overall, this dataset provides a strong foundation for understanding how **daily habits influence long-term health outcomes**, while remaining flexible for data analysis and visualization.

3. Data Analysis using SQL

1 Does working more hours lead to shorter or longer lifespans?

	avg_work_hours	avg_lifespan
▶	7.82	83.80
	11.19	76.68
	3.47	75.91

2 What is the optimal amount of sleep for maximum longevity?

	sleep_hours	avg_lifespan
▶	2.1	85.00
	7	84.15
	7.5	84.05
	4.7	84.05
	7.8	84.01

3 How does exercise impact lifespan (controlled view)?

	exercise_hours	avg_lifespan
▶	0	68.93
	0.1	78.25
	0.2	76.83
	0.3	77.75
	0.4	78.73
	0.5	78.09
	0.6	78.88

4 Are there occupation-specific patterns in longevity?

	occupation_type	avg_lifespan
▶	Teacher	83.19
	Artist	83.19
	Retail Worker	82.46
	Freelancer	81.15
	Office Worker	80.86
	Technician	80.63
	Engineer	80.48

Q5 Do extreme outliers (24-hour work days) affect results?

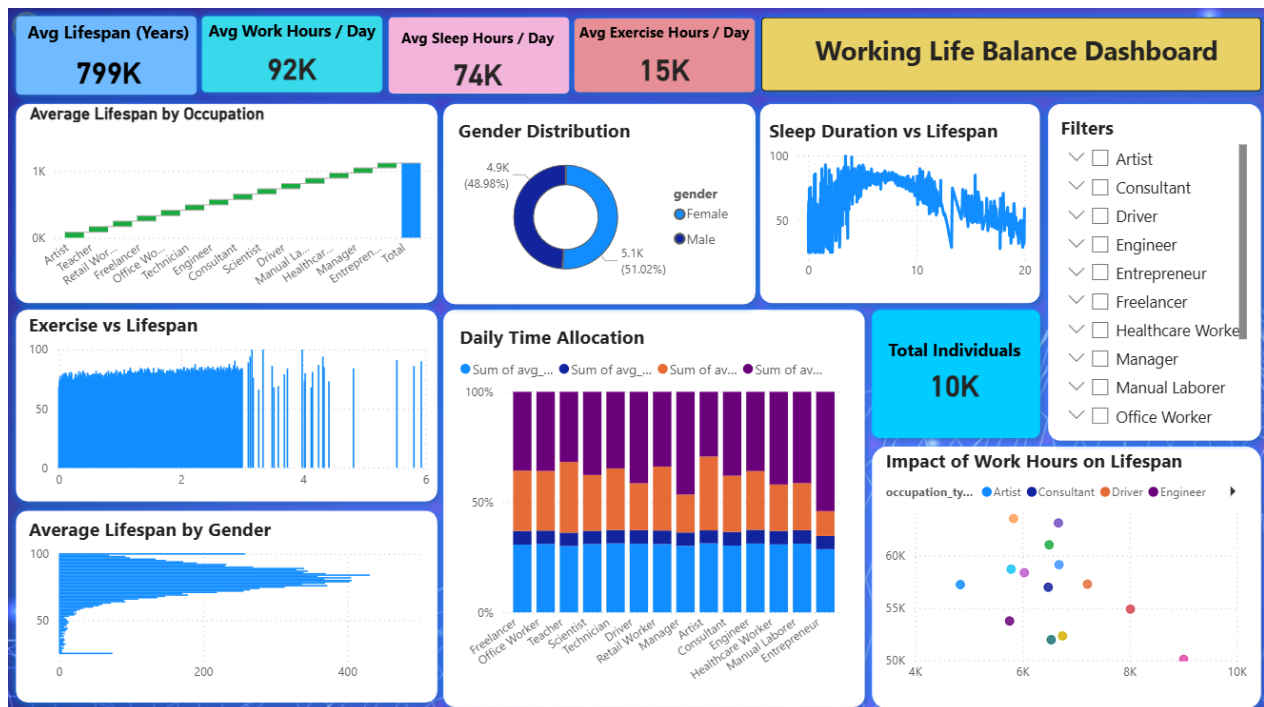
	extreme_workers	avg_lifespan
▶	131	29.82

Q6 Is there a gender gap in longevity?

	gender	avg_lifespan
▶	Female	81.36
	Male	78.28

5.dashboard in power Bi

Finally we built an interactive dashboard in power bi to present insight visually



Recommendations

Based on the analysis presented in the **Working Life Balance Dashboard**, the following recommendations are suggested to improve overall health, productivity, and longevity:

1. Maintain Balanced Work Hours

Individuals working **moderate daily hours** tend to have a higher lifespan compared to those who overwork.

Organizations should encourage **reasonable working hours** and avoid prolonged work schedules to reduce stress and health risks.

2. Promote Healthy Sleep Patterns

The analysis shows that **consistent and adequate sleep (around 7–8 hours per day)** is strongly associated with better longevity.

Employers and individuals should prioritize sleep-friendly schedules and reduce habits that disrupt rest.

3. Encourage Regular Physical Activity

Moderate daily exercise has a **positive impact on lifespan**.

Simple activities such as walking, stretching, or light workouts should be integrated into daily routines, especially for sedentary professions.

4. Support Work-Life Balance Across Occupations

Different occupations show varied daily time allocation patterns.

Organizations can use these insights to:

- Design healthier work policies
- Reduce burnout in high-stress roles
- Improve employee well-being and retention

5. Use Data-Driven Lifestyle Monitoring

Interactive dashboards like this can help decision-makers:

- Identify unhealthy lifestyle trends
- Compare groups using filters (occupation, gender, age)
- Take preventive actions based on data insights

6. Focus on Overall Lifestyle Balance

Longevity is influenced by a **combination of factors**, not a single habit.

A balanced approach that includes **work moderation, quality sleep, regular exercise, and sufficient rest** leads to healthier long-term outcomes.